

Domain	Cooling
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code	service	✓1						
C-1a	Cooling emission control	Service group: Cooling control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Central automatic control	1	0	1	1	1	0	0
level 2	Individual room control	1	0	1	2	2	0	0
level 3	Individual room control with communication between controllers and to BACS	2	0	2	3	2	1	0
level 4	Individual room control with communication and occupancy detection	3	0	2	3	2	1	0
Information sources Standard?								
		EN 15232						

code	service	✖0						
C-1b	Emission control for TABS (cooling mode)	Service group: Cooling control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Central automatic control	1	0	1	1	1	0	0
level 2	Advanced central automatic control	1	0	1	1	2	0	0
level 3	Advanced central automatic control with intermittent operation and/or room temperature feedback control	2	0	2	3	2	1	1
level 4	0							
Information sources								
		Standard?	EN 15232					

code	service	
C-1c	Control of distribution network chilled water temperature (supply or return)	Service group: Cooling control - demand side

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Constant temperature control	0	0	0	0	0	0	0
level 1	Outside temperature compensated control	1	0	1	1	0	0	0
level 2	Demand based control	2	0	1	1	0	0	0
level 3	0	-	-	-	-	-	-	-
level 4	0							
Information sources								
		Standard?	EN 15232					

code	service	✗0
C-1d	Control of distribution pumps in networks	Service group: Cooling control - demand side

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	On off control	1	0	0	0	0	0	0
level 2	Multi-Stage control	2	0	0	0	0	0	0
level 3	Variable speed pump control (pump unit (internal) estimations)	2	0	0	0	0	0	0
level 4	Variable speed pump control (external demand signal)	2	0	0	0	0	0	0
Information sources								
		Standard?	EN 15232					

code	service	✗0
C-1f	Interlock: avoiding simultaneous heating and cooling in the same room	Service group: Cooling control - demand side

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No interlock	0	0	0	0	0	0	0

level 1	Partial interlock (minimising risk of simultaneous heating and cooling e.g. by sliding setpoints)	2	0	0	0	0	0	0
level 2	Total interlock (control system ensures no simultaneous heating and cooling can take place)	3	0	0	0	0	0	0
level 3	0	-	-	-	-	-	-	-
level 4	0							
Information sources								
		Standard?	EN 15232					
code	service	0						
C-1g	Control of Thermal Energy Storage (TES) operation	Service group: Cooling control - demand side						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Continuous storage operation	0	0	0	0	0	0	0
level 1	Time-scheduled storage operation	1	0	0	0	0	0	0
level 2	Load prediction based storage operation	2	1	0	0	0	0	0
level 3	Cold storage capable of flexible control through grid signals (e.g. DSM)	2	2	0	0	0	0	0
level 4	0							
Information sources								
		Standard?	EN 15232					
code	service	1						
C-2a	Generator control for cooling	Service group: Control cooling production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	On/Off-control of cooling production	0	0	0	0	0	0	0

level 1	Multi-stage control of cooling production capacity depending on the load or demand (e.g. on/off of several compressors)	1	1	1	0	0	0	0
level 2	Variable control of cooling production capacity depending on the load or demand (e.g. hot gas bypass, inverter frequency control)	2	1	2	0	0	0	0
level 3	Variable control of cooling production capacity depending on the load AND external signals from grid	2	3	2	0	0	0	0
level 4	0							
Information sources								
Standard?		EN 15232						
code	service	✗0						
C-2b	Sequencing of different cooling generators	Service group: Control cooling production facilities						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Priorities only based on running times	0	0	0	0	0	0	0
level 1	Fixed sequencing based on loads only: e.g. depending on the generators characteristics such as absorption chiller vs. centrifugal chiller	1	0	0	0	0	0	0
level 2	Dynamic priorities based on generator efficiency and characteristics (e.g. availability of free cooling)	2	1	0	0	0	0	0
level 3	Load prediction based sequencing: the sequence is based on e.g. COP and available power of a device and the predicted required power	3	2	0	0	0	0	0
level 4	Sequencing based on dynamic priority list, including external signals from grid	3	3	0	0	0	0	0
Information sources								
Standard?		EN 15232						
code	service	✔1						
C-3	Report information regarding cooling system performance	Service group: Information to occupants and facility managers						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0

level 1	Central or remote reporting of current performance KPIs (e.g. temperatures, submetering energy usage)	1	0	0	0	0	1	1
level 2	Central or remote reporting of current performance KPIs and historical data	1	0	0	0	0	1	2
level 3	Central or remote reporting of performance evaluation including forecasting and/or benchmarking	1	0	0	0	0	1	3
level 4	Central or remote reporting of performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	1	0	0	1	0	3	3
Information sources								
		Standard?						
code	service	1						
C-4	Flexibility and grid interaction	Service group: Flexibility and grid interaction						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Scheduled operation of cooling system	1	0	1	1	0	0	0
level 2	Self-learning optimal control of cooling system	2	1	2	2	0	0	0
level 3	Cooling system capable of flexible control through grid signals (e.g. DSM)	2	3	2	3	0	0	0
level 4	Optimized control of cooling system based on local predictions and grid signals (e.g. through model predictive control)	2	3	3	3	1	0	0
Information sources								
		Standard?						